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Assignment -1

Q1 Write a program to demonstrate InetAddress class

```
import java.net.*;  
public class NetDemo2  
{  
    public static void main(String[] args) {  
        try {  
            InetAddress i = InetAddress.getLocalHost();  
            System.out.println("Local Address:" + i);  
  
            InetAddress i1 = InetAddress.getByName  
                ("www.google.com");  
            System.out.println("Host Address:" + i1);  
            System.out.println("Host Name:" + i1.getHostName());  
        }  
        catch (Exception e)  
        {  
            System.out.println(e);  
        }  
    }  
}
```

Output :-

```
Local Address : DESKTOP-DTGKJBG/169.254.39.121  
Host Address : www.google.com/142.251.42.100  
Host Name : www.google.com
```

Q9. Write a program to demonstrate socket class.

```
import java.net.*
import java.io.*
public class Demo
{
    public static void main (String [] args)
    {
        try {
            Socket s = new Socket ("www.google.com", 80);
            System.out.println ("Connected to" + s.getInetAddress()
            + s.getPort () + "from port " + s.getLocalPort () + "of" +
            s.getLocalAddress());
        }
        catch (Exception e)
        {
            System.out.println (e);
        }
    }
}
```

Output :-

Connected to www.google.com/172.217.27.196 80 from
port 53771 of 192.167.1.4

Q3 Write a program to demonstrate URL class.

```
import java.net.*;  
public class URLEdemo  
{  
    public static void main (String[] args)  
    {  
        try{  
            URL url = new URL("http://www.yahoo.com:80/index");  
  
            System.out.println("Host Name : " + url.getHost());  
            System.out.println(" port : " + url.getPort());  
        }  
        catch (Exception e)  
        {  
            System.out.println(e);  
        }  
    }  
}
```

Output :-

Host Name : www.yahoo.com

Port : 80

Q4 Write a program to insert and update record in student table using JDBC driver

```
import java.sql.*;

public class abc {
    public static void main (String [] args) throws
        ClassNotFoundException, SQLException
    {
        Class.forName ("oracle.jdbc.driver.OracleDriver");
        Connection con = DriverManager.getConnection ("jdbc:oracle:
            thin:1521:xe", "system", "abc@123");

        System.out.println ("Connection to Database created");
        Statement sta = con.createStatement();
        sta.executeUpdate ("insert into student values (1, 'Ram',
            'Pune')");
        System.out.println ("Record Inserted Successfully");
        con.close();
    }
}
```

Output :-

Connection to Database created
Record Inserted Successfully

Update data from student table

```
import java.sql.*;

public class abc {

    public static void main (String [] args) throws
        ClassNotFoundException, SQLException
    {
        Class.forName ("oracle.jdbc.driver.OracleDriver");
        Connection con = DriverManager.getConnection (
            "jdbc:oracle:thin:@localhost:1521:xe", "system", "abc@123");

        System.out.println ("Connection to database created");
        Statement sta = con.createStatement ();

        Int result = sta.executeUpdate ("update student
            set name = 'Raj' where rollno = 5");

        System.out.println (result + " record updated");
        con.close();
    }
}
```

Output:-

Connection to database created

Assignment -2

Q1. Write a program by using GenericServlet.

```
import java.io.*;
import java.servlet.*;
public class First extends GenericServlet
{
    public void service (ServletRequest req, ServletResponse res)
        throws IOException, ServletException
    {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.print("<html> <body>");
        out.print(" <b> Hello Generic servlet </b>");
        out.print(" </body></html>");
    }
}
```

Write a program by using HttpServlet.

```
import javax.servlet.http.*;
import javax.servlet.*;
import java.io.*;

public class Demoservlet extends HttpServlet
{
    public void doGet (HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException
    {
        res.setContentType("text/html");
        PrintWriter pw = res.getWriter();

        pw.println("<html> <body>");

        pw.println("<b> Welcome to servlet </b>");
        pw.println("</body> </html>");

        pw.close();
    }
}
```

Write a program to demonstrate cookies.

index.html

```
<form action = "servlet1" method = "post">  
Name : <input type = "text" name = "userName" /> <br>  
<input type = "submit" value = "go" />  
</form>
```

FirstServlet.java

```
import java.io.*;  
import javax.servlet.*;  
import javax.servlet.http.*;  
public class FirstServlet extends HttpServlet {  
    public void doPost (HttpServletRequest req, HttpServletResponse res) {  
        try {  
            res.setContentType ("text/html");  
            PrintWriter out = res.getWriter ();  
  
            String n = req.getParameter ("userName");  
            out.print ("Welcome " + n);  
            Cookie ck = new Cookie ("uname", n);  
            res.addCookie (ck);  
  
            out.print ("<form action = 'servlet2'>");  
            out.print ("<input type = 'submit' value = 'go'>");  
            out.print ("</form>");  
            out.close ();  
        }  
        catch (Exception e)  
        {  
            System.out.println (e);  
        }  
    }  
}
```

```

secondServlet.java
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class secondServlet extends HttpServlet
{
    public void doPost (HttpServletRequest req, HttpServletResponse res)
    {
        try {
            res.setContentType("text/html");
            PrintWriter out = res.getWriter();

            Cookie ck[] = req.getCookies();
            out.print("Hello " + ck[0].getValue());

            out.close();
        }
        catch (Exception e)
        {
            System.out.println(e);
        }
    }
}

```

Web.xml

```

<web-app>
  <servlet>
    <servlet-name> s1 </servlet-name>
    <servlet-class> FirstServlet </servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name> s1 </servlet-name>
    <url-pattern> /servlet1 </url-pattern>
  </servlet-mapping>

  <servlet>
    <servlet-name> s2 </servlet-name>
    <servlet-class> SecondServlet </servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name> s2 </servlet-name>
    <url-pattern> /servlet2 </url-pattern>
  </servlet-mapping>
</web-app>

```

Output :-

← → C ↻ localhost:8888/cookies/ ↻

Name :

← → ↻ localhost:8888/cookies/servlet1 ↻

Welcome Ram

← → ↻ localhost:8888/cookies/servlet2 ↻

Hello Ram

Write a program to demonstrate request dispatcher using forward method

index.html

```
<form action = "servlet1" method = "post">
Name: <input type = "text" name = "userName" /> <br />
Password: <input type = "password" name = "userPass" /> <br />
<input type = "submit" value = "login" />
</form>
```

Login.java

```
import java.io.*;
import java.servlet.*;
import java.servlet.http.*;

public class Login extends HttpServlet
{
    public void doPost (HttpServletRequest req, HttpServletResponse res)
    { throws ServletException, IOException
        {
            res.setContentType ("text/html");
            PrintWriter out = res.getWriter ();
            String n = req.getParameter ("userName");
            String p = req.getParameter ("userPass");
            if (p.equals ("paniv")
            {
                RequestDispatcher rd = req.getRequestDispatcher ("servlet2");
                rd.forward (req, res);
            }
            else
            {
                out.print ("Sorry Username or Password Error!");
                RequestDispatcher rd = req.getRequestDispatcher ("/index.html");
                rd.include (req, res);
            }
        }
    }
}
```

```

WelcomeServlet.java
import java.io.*
import javax.servlet.*;
import javax.servlet.http.*;

public class WelcomeServlet extends HttpServlet {

    public void doPost(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException {

        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        String n = req.getParameter("username");
        out.print("Welcome" + n);

    }
}

```

Web.xml

```

<web-app>
  <servlet>
    <servlet-name> Login </servlet-name>
    <servlet-class> Login </servlet-class>
  </servlet>
  <servlet>
    <servlet-name> WelcomeServlet </servlet-name>
    <servlet-class> WelcomeServlet </servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name> Login </servlet-name>
    <url-pattern> /servlet1 </url-pattern>
  </servlet-mapping>
  <servlet-mapping>
    <servlet-name> Welcome </servlet-name>
    <url-pattern> /servlet2 </url-pattern>
  </servlet-mapping>
  <welcome-file-list>
    <welcome-file> index.html </welcome-file>
  </welcome-file-list>
</web-app>

```

Input :-

When Entering user name or password incorrect.

← → ↻ ☆

Name :

Password :

← → ↻

Sorry username or password error !

Name :

Password :

← → ↻

Welcome SIIIT

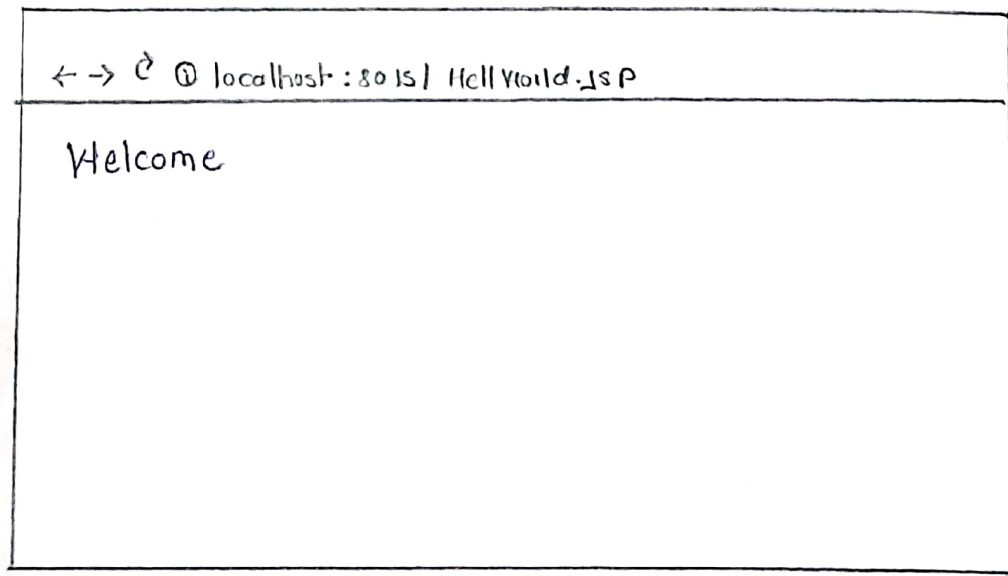
Assignment-3

Q1. Write a program to display 'Welcome' message in jsp.

```
<%@page language="java" contentType="text/html" charset=
    ISO-8859-1" pageEncoding="ISO-8859-1"%>
```

```
<html>
<head>
<title> Declaration </title>
</head>
<body>
    <h1> Welcome </h1>
</body>
</html>
```

Output :-



Q2. Write a JSP program which will use the JSTL coreTags.

```
<%@ taglib uri = "http://java.sun.com/jsp/jstl/core" prefix = "c"%>
<!doctype html>
<html>
<body>
<h2> Using JSTL core Tags </h2>
<c:set var = "message" value = "Welcome to JSTL in JSP!" />
<p> Message : <c:out value = "${message}" /> </p>
<c:set var = "showMessage" value = "true" />
<c:if test = "${showMessage}">
    <p> This message is displayed because showMessage is true. </p>
</c:if>
<c:set var = "fruits" value = "${'Apple', 'Banana', 'cherry'}" />
<h3> Fruit List : </h3>
<ul>
    <c:forEach var = "fruit" items = "${fruits}">
        <li> <c:out value = "${fruit}" /> </li>
    </c:forEach>
</ul>
</body>
</html>
```

Output :- Using JSTL core Tags
Message : Welcome to JSTL in JSP!
This message is displayed because showMessage is true
Fruit List :

- Apple
- Banana
- cherry

Q3 Write a program to demonstrate request & response implicit object in JSP.

* Request object

index.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<form action = "Welcome.jsp">
```

```
<input type="text" name = "uname"> <input type="submit" value = "go"> <br/>
```

```
</form> </body> </html>
```

Welcome.jsp

```
<%@ page language = "Java" contentType = "text/html" ;  
charset = "ISO-8859-1" pageEncoding = "ISO-8859-1" %>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<%
```

```
String name = request.getParameter("uname");
```

```
out.print("Welcome" + name);
```

```
%>
```

```
</body>
```

```
</html>
```

Output:-

← → @localhost:8015/HelloworldJSP/index.jsp

Type your Name and click go button.

← → @localhost:8015/HelloworldJSP/welcome.jsp?username=Ram

Welcome Ram

* Response Object

```
index.html
<!DOCTYPE html>
<html>
<body>
<form action = "Welcome.jsp">
<input type = "text" name = "uname" />
<input type = "submit" value = "Go" /> <br />
</form>
</body>
</html>
```

Welcome.jsp

```
<% @ page language = "java" content type = "text/html ; charset = ISO-8859-1" %>
pageEncoding = "ISO-8859-1" %>
<!DOCTYPE html>
<html> <body>
<%
    response.sendRedirect("http://www.google.com");
%>
</body> </html>
```

Output :-

← → ↻ @localhost:8015/HelloWorldJSP/index.html

Type your name & click "Go" Button, you will be redirected to "Google.com".

google.com

Google